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United States Patent Application Publication

20250281504

Kind Code

A1

Publication Date

September 11, 2025

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Method for improving insulin insensitivity

Abstract

A method of ameliorating a disorder of insulin sensitivity, which entails administering an amount of one or more glucocorticoid receptor antagonists effective to ameliorate the disorder with no more than an acceptable activation of the hypothalamic-pituitary-adrenal axis, such that 24 hour serum or urine cortisol levels do not exceed about two to three times the upper normal limit, respectively.

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Family ID: 1000008626536

Appl. No.: 18/831326

Filed: November 19, 2024

Related U.S. Application Data

parent US continuation 17438580 20210913 ABANDONED US continuation PCT/US20/00009
20200309 child US 18831326
us-provisional-application US 62919496 20190318

Publication Classification

Int. Cl.: A61K31/567 (20060101); A61K31/4745 (20060101); A61K31/513 (20060101);
A61K31/57 (20060101); A61K45/06 (20060101); A61P3/08 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

[0001] Insulin sensitivity is a measure of the degree to which cells respond to insulin. The converse, insulin insensitivity, (also termed insulin resistance), is a pathological condition wherein the cellular response to the presence of insulin in the bloodstream is diminished to the extent that homeostasis cannot be maintained and the normal metabolism of glucose is disrupted.

[0002] Normally, insulin causes cells to take up glucose, which is used as an energy source. Thus, insulin is a critical part of the process by which normal levels of blood glucose are maintained. If cells become less sensitive to insulin, that is, insulin resistant, they become less effective in taking up glucose and blood glucose levels rise beyond normal levels. This reduced activity of insulin may cause prediabetes, diabetes, metabolic syndrome, non-alcoholic fatty liver disease and other diseases of metabolism.

[0003] Insulin sensitivity and resistance may occur differentially in the various tissues that are involved in glucose and fat metabolism, including, but not limited to, the liver, skeletal muscles and adipose tissues. Each of these contributes to “total” or overall insulin insensitivity and resistance, and prolonged changes in hepatic and adipose insulin sensitivity lead to changes in total body insulin sensitivity.

[0004] Glucocorticoids include corticosteroid hormones produced in the adrenal cortex, as well as synthetic analogs of these hormones. In addition to other physiologic activities, glucocorticoids are involved in carbohydrate (e.g., glucose) fat and protein metabolism. Cortisol is the main glucocorticoid in humans.

[0005] Among other actions, glucocorticoids have been shown to stimulate gluconeogenesis, mobilize amino acids, and inhibit glucose uptake in muscle and fat. Chronically increased blood levels of the glucocorticoid, cortisol, occur in Cushing's Syndrome. The increased glucocorticoid action in this condition is implicated in abnormal glucose and fat metabolism leading to obesity, metabolic syndrome and diabetes.

[0006] Glucocorticoids exert their actions by binding to the glucocorticoid receptor, which is a ligand activated nuclear receptor. The ligand-bound-receptor translocates to the nucleus where it binds to specific sites on DNA, leading to RNA transcription and then protein synthesis. It has been shown that the effects resulting from glucocorticoid binding to the glucocorticoid receptor can be inhibited at least partially by the administration of anti-glucocorticoids, that is, synthetic molecules that bind to the glucocorticoid receptor and prevent it from acting on DNA to induce transcription.

[0007] Appropriate levels of cortisol are critical for human homeostasis and normal metabolism, which may explain the tight regulation of its production. Cortisol biosynthesis and secretion are regulated by adrenocorticotrophic hormone (ACTH) produced in the pituitary gland, which in turn is stimulated by corticotropin-releasing hormone (CRH), produced and secreted by the hypothalamus. Cortisol and ACTH levels (and by inference CRH levels) are lowest after the onset of sleep, gradually rising before waking and falling beginning a few hours thereafter. This circadian pattern appears to be critical to health. While the factors that activate the system are not fully understood, it is clear that increasing cortisol levels have a “negative feedback” effect to inhibit ACTH secretion, so that cortisol partially governs its own rate of secretion.

[0008] Currently, the number of people who are characterized as being overweight and/or obese is increasing dramatically. This represents a major healthcare challenge due also to the associated co-morbidities of hypertension, hyper-lipidemia and glucose intolerance/diabetes. Hence, there is an

unmet and critical need to reverse elements of metabolic syndrome, including hypertension, weight gain, hyperlipidemia and glucose intolerance/diabetes as pre-diabetes (pre-DM) and overt diabetes mellitus (DM) continue to exert substantial physical and financial burden on patients worldwide. [0009] The estimated annual incidence of diabetes in the United States population is projected to increase from 8 cases per 1,000 in 2008 to 15 cases per 1,000 by 2050, and the prevalence is projected to rise from 14% in 2010 to 25-28% by 2050. Uncontrolled diabetes accounts for about 10% of deaths, in part from its comorbidities of cardiovascular disease and renal failure. In view of these current statistics and projections, provision of novel treatments to improve glucose tolerance and diabetes are urgently needed to prevent progression of pre-diabetes and to reduce the morbidity and mortality of diabetes.

SUMMARY OF THE INVENTION

[0010] Accordingly, it is an object of the present invention to provide a method of ameliorating a disorder of insulin sensitivity by administration of an amount of one or more glucocorticoid receptor antagonists and/or inhibitors which amount does not activate the hypothalamic-pituitary-adrenal (HPA) axis, or increases daily urine cortisol secretion no more than about three times the upper limit of the normal range, or increases 24-hour mean serum cortisol or area under the curve (AUC) no more than twice normal.

[0011] It is also an object of the present invention to provide a method of ameliorating a disorder associated with insulin insensitivity, such as diabetes, prediabetes, overweight/obesity, lipodystrophy, hypertriglyceridemia, and/or growth hormone excess states. It is also an object of the present invention to improve insulin sensitivity in individuals who are neither overweight nor obese.

[0012] It is further an object of the present invention to provide a method for improving insulin sensitivity of a patient by blocking or otherwise inhibiting glucocorticoid receptors over an extended period of time where activation of the hypothalamic-pituitary-adrenal axis is preferably minimized.

[0013] However, given the exigencies of life, transient periods of stress, depression and pseudo-Cushing's Syndrome can cause a subject's urinary free cortisol to increase for a limited period of time. Thus, it is within the scope of the present invention to accommodate urinary free cortisol levels preferably less than, or equal to, twice normal, but no more than three times the upper limit of normal. In fact, up to three times normal urinary free cortisol is what is observed in pseudo-Cushing's Syndrome.

[0014] It is, moreover, an object of the present invention to provide a method of treating insulin sensitivity in a patient by at least partially blocking glucocorticoid receptors in tissues responsible for glucose and fat metabolism.

[0015] It is, moreover, an object of the present invention to provide a method of improving insulin sensitivity in people having normal cortisol regulation.

[0016] It is further an object of the present invention to provide a method of improving insulin sensitivity by administering low doses of GR antagonists more than once per day to improve insulin sensitivity without inducing hypercortisolism.

[0017] It is, additionally, an object of the present invention to improve insulin sensitivity in hepatic and adipose tissue, in particular.

[0018] It is further an object of the present invention to use assays for measuring urine cortisol levels or other surrogate biomarkers of cortisol to determine if dosages of administered GR antagonists and/or inhibitors need to be adjusted downward to prevent 24-hour serum or urine cortisol levels from exceeding about two to three times the upper limit of normal, respectively.

[0019] It is, moreover, an object of the present invention to provide a method of improving insulin resistance in humans in need thereof under non-rescue conditions.

[0020] The above objects and others discussed below are provided by a method of ameliorating an insulin insensitivity state in a human patient, which entails administering an amount of

glucocorticoid receptor antagonist, which amount maintains 24-hour urine or serum cortisol levels within the normal range or up to three or two-fold normal, respectively, in the human patient.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 illustrates the major metabolic pathways of mifepristone in vivo, including Mono-demethylated mifepristone (metapristone), C-hydroxylated mifepristone, and Di-demethylated mifepristone, these metabolites being formed by oxidation as shown in this figure by cytochromes P450 3A4 (CYP3A4) and P450 3A5 (CYP3A5).

[0022] FIG. 2 illustrates the urine cortisol excretion over 24 hours at baseline (before administration of study agent), and at the end of administration of a placebo or mifepristone (at a dose of 50 mg every six hours for nine days). The upper limit of the normal reference range is 45 micrograms/day.

[0023] FIG. 3 is a flow diagram of participants throughout a study described herein.

[0024] FIG. 4 illustrates a 24-hour profile of serum ACTH and cortisol in study participants at the end of either placebo or mifepristone administration.

[0025] FIG. 5 illustrates time profiles of glucose and insulin following an oral glucose tolerance test (OGTT) in study participants at the end of either placebo or mifepristone administration.

[0026] FIG. 6 illustrates the effects of placebo and mifepristone on indices of insulin insensitivity.

[0027] FIG. 7 illustrates the effects of placebo and mifepristone on insulin secretory rates and insulin clearance.

[0028] FIG. 8 illustrates the effects of placebo and mifepristone on pancreatic β -cell responsivity indices.

DETAILED DESCRIPTION OF THE INVENTION AND EMBODIMENTS

[0029] The present invention is based, at least in part, on the discovery that administration of mifepristone or any of its metabolites or derivatives to a human patient in specific dosage ranges and/or at specific times of the day will reverse elements of the metabolic syndrome.

[0030] The present invention is also based, at least in part, upon the discovery that administration of mifepristone or any of its metabolites or derivatives to a human patient in specific dosage ranges and/or at specific times of the day will facilitate weight loss.

[0031] The present invention is also based, at least, in part, upon the discovery that administration of one of more GR antagonists, steroidal or non-steroidal, in low doses and more than once per day improves insulin sensitivity without inducing biochemical hypercortisolism in humans. The present invention may be viewed conversely as decreasing insulin resistance in humans in need thereof.

[0032] The present invention also provides a method of improving insulin sensitivity by administering an amount of glucocorticoid receptor antagonists to a human patient in an amount that does not increase mean 24-hour serum cortisol or 24-hour urine cortisol levels more than about two or three times the upper limit of the normal range, respectively, over an extended period of time, such as several weeks or several months.

[0033] The present invention also provides a method for at least partially blocking, in a human patient, glucocorticoid receptors in tissues that are responsible for glucose and fat metabolism, and to do so over an extended period of time without stimulating the hypothalamic-pituitary-adrenal axis, or with stimulation of 24-hour serum or urine cortisol no more than two or three times the upper limit of normal, respectively.

Term Definitions

[0034] The following terms used throughout the present specification are defined as follows:

[0035] About: means plus or minus 10% of any value, number or quantity for which the term is applied. For example, as defined here, the phrase “from about 10 to 100” means from 9 to 110.

Thus, term “about” as defined herein specifically applies to ranges of values, numbers or quantities. [0036] Mifepristone: is 11beta-[p-(dimethylamino) phenyl]-17alpha-(1-propynyl) estr-4,9-dien-17beta-ol-3-one, also known as RU-486 or RU-38486. The term “analog” of mifepristone is defined herein as substituents at the 11 and/or 17-positions of the mifepristone ring system that increase GR antagonism by increased selectivity for the GR receptors. Examples of such substituents are the addition of a t-butyl group to the acetylenic group at C-17, or one or more fluorine substituents on the dimethylamino group at C-11, or replacement of the dimethylamino group on the phenyl ring at C-11 with heterocyclic rings, including, for example, morpholine, azolidine, oxacyclobutanes which are optionally methyl-substituted at the steroid ring attachment position and thiomorpholine or sulfone thiomorpholine. Further, the analogs of mifepristone as defined in the present invention satisfy the following conditions: GR antagonism: IC₅₀ (nM); <20; GR agonism: BC_{sub.50} (nM) [E_{max}]: >2,500; and AR agonism: BC₅₀ (nM) [E_{max}] >2,500. See Rew Y et al. *J. Med. Chem.* 2018; 61, 7767-7784. [0037] Metabolites of mifepristone: means common in vivo metabolites, including for example, mono-demethylated mifepristone (also known as RU-42633, as well as metapristone), N-didemethyl mifepristone (also known as RU-42848) and hydroxylated mifepristone (also known as RU-42698). Further, by reference to the attached drawing, it is noted here that prodrug modifications or analog modifications may be used to prepare structural modifications of mifepristone shown in the drawing. For example, the hydroxy groups shown in these four structures may be etherified, and the etherified compounds used in the present invention. Specifically, acetyl esters are preferred. This is an example of a prodrug. Also, various lower alkyl and lower dialkyl groups may be used as substituent or replacement groups on the dimethylamino group of mifepristone or for the respective amino, monomethylamino or dimethylamino of any of the three major metabolites thereof, for example. By “alkyl” is meant C_{sub.1}-C_{sub.6}, including methyl, ethyl, propyl, butyl, pentyl and hexyl groups. Thus, for example, diethylamino-mifepristone is an exemplary compound. These analog or prodrug compounds may be prepared using well-known chemical reactions. [0038] Antigluccorticoids or glucocorticoid antagonists: means molecules or compounds that reduce the action of glucocorticoids at their receptors by blocking those receptors with the effect of inhibiting, preventing, blockading or antagonizing glucocorticoid action. In accordance with the present invention any glucocorticoid antagonist may be used. These antagonists may include, for example, mifepristone and any of its major metabolites in vivo. However, other compounds that also function as glucocorticoid receptor antagonists may be used instead. Glucocorticoid receptor antagonists are also referred to as GR antagonists or glucocorticoid (GR) inhibitors or blockers. The GR antagonists contemplated herein may be either steroidal or non-steroidal compounds. However, it is preferable that these compounds exhibit minimal activity for the progesterin receptor. GR antagonists are also referred to herein as GRAs.

[0039] Any of the GR antagonist and/or inhibitor compounds disclosed in U.S. Pat. Nos. 6,362,173; 6,964,953; 5,064,822; 10,238,659; 6,011,025 may be used in conjunction with the present invention. Each and all of these patents are incorporated by reference herein in the entirety. However, the compounds contemplated by the present invention do not block or interfere with cortisol synthesis, itself. Also incorporated herein by reference in the entirety are US published patent applications 2019/151313 and 2016/0289261.

[0040] Further, in accordance with the present invention either steroidal or non-steroidal GRAs may be used. Examples of steroidal GRAs that may be used are those of any of U.S. Pat. Nos. 4,296,206; 4,386,085; 4,447,424; 4,477,445; 4,519,946; 4,540,686; 4,547,493; 4,634,695; 4,634,696; 4,753,932; 4,774,236; 4,808,710; 4,814,327; 4,829,060; 4,861,763; 4,912,097; 4,921,638; 4,943,566; 4,954,490; 4,978,657; 5,696,127; 6,303,591; and 10,151,763; each and all of which are incorporated herein in the entirety by reference. Examples of non-steroidal GRAs which may be used are those of any of U.S. Pat. Nos. 5,696,127; 6,570,020; 6,051,573 and 10,151,763; each and all of which are incorporated herein by reference in the entirety.

[0041] Further identification of additional GR antagonists may be determined by one of ordinary skill in the art by using known kinetic and functional assays to distinguish between GR agonists and GR antagonists. For example, see Clark R D, *Curr. Top. Med. Chem.* 2008; 8(9): 813-838, "Glucocorticoid Receptor Antagonists", which neatly summarizes recent progress in the discovery of selective glucocorticoid GR antagonists.

[0042] Various known methods for evaluating GR antagonist properties, including binding, functional and in vivo assays are disclosed. Further, structure-activity relationships (SAR) of non-steroidal GR antagonists from numerous structural classes, such as octahydrophenanthrenes, spirocyclic dihydropyridines, triphenylmethanes, diaryl ethers, chromenes, dibenzyl anilines and dihydroiso-quinolines are presented. Hence, one skilled in the art is able to determine and verify whether a compound exhibits predominant GR antagonism using known assays.

[0043] Further, high-throughput screening (HTS) may be used to rapidly screen large numbers of compounds to not only identify GR antagonists, but also to determine and verify that identified GR antagonists have minimal GR agonist activity.

[0044] However, of particular use, mention may be made of GR luciferase antagonism assays. See Rew et al., *Journal of Medicinal Chemistry* 2018, (1, 7767-7784). Luciferase assay kits are commercially available for conducting these assays. For example, mention may be made of the dual-luciferase reporter assay for nuclear receptors of Promega Corporation of Madison, Wisconsin or the GR reporter assay system of Indigo Biosciences of State College, Pennsylvania. Often, the GR antagonism activity is measured against mifepristone (RU 486). Further, using mifepristone as a starting compound, compounds having selectively improved GR antagonist activity can be readily identified based on a quantitative structure activity relationship (QSAR). In fact, it is known that the C11 and C17 positions on the mifepristone ring system are sensitive to structural modifications in altering GR antagonism as are the bonding between C0 and C10. See the carbon-numbered molecular structure of mifepristone at the present specification in Structural Appendix A.

[0045] The guiding principle is to select substituents at C11 and C17 as shown in structure A (mifepristone) which improve GR antagonism, while minimizing GR agonism. Compounds that meet these criteria are shown in *J. Med. Chem.*, supra, such as compounds 11, 18, 22-25, 27 and 28 in Tables 1, 2 and 3 thereof. Of particular importance as a selective GR antagonist is the compound shown on the right-hand side of structures B, i.e., B2. The compound on the left-hand side of B, i.e., B1, is mifepristone. Structures B1 and B2 are shown in the Structural Appendix of the present specification. In fact, it has been known since at least 1989 that the two most commonly known classes of structural modifications of the cortisol steroid backbone for obtaining GRAs include: 1) modification of the 11- β hydroxy group, and 2) modification of the 17- β side chain. See Lefebvre, *J. Steroid Biochem.* 33:557-563 (1989).

[0046] Some preferred examples of non-steroidal GRAs that may be used in the present invention include GRAs having a cyclohexyl-pyrimidine backbone as disclosed U.S. Pat. No. 10,151,763, incorporated by reference herein in the entirety. Included in this class of compounds is the compound Miricorilant, whose IUPAC name is 6-(4-phenylcyclohexyl)-5-[[3-(trifluoromethyl)phenyl]methyl]-1H-pyrimidine-2,4-dione. The molecular structure of this compound is shown in Structural Appendix D of the present specification.

[0047] Other preferred examples of non-steroidal GRAs that may be used in the present invention include GRAs having a fused azadecalin backbone as disclosed in U.S. Pat. Nos. 7,928,237 and 10,151,763, both of which are incorporated herein in the entirety. Included in this class of compounds is the compound Relacorilant, whose IUPAC name is [(4aR)-1-(4-fluorophenyl)-6-(1-methylpyrazol-4-yl) sulfonyl-4,5,7,8-tetrahydropyrazolol[3,4-g] isquinolin-4a-yl]-[4-(trifluoromethyl)pyridin-2-yl]methanone. The molecular structure of this compound is shown in Structural Appendix C of the present specification. This compound may also be purchased commercially.

[0048] Another example of a non-steroidal GR antagonist is Exicorilant, the IUPAC name of which

is ((4aR,8aS)-1-(4-fluorophenyl)-1,4,5,6,7,8,8a,9-octahydro-6-((2-methyl-2H-1,2,3-triazol-4-yl)sulfonyl)-4ah-pyrazolo (3,4-g)isoquinolin-4a-yl)(4-(trifluoromethyl)-2-pyridinyl)-methanone, and shown in Structural Appendix E.

[0049] Further, assays for detecting GR antagonists are described in U.S. Pat. No. 8,940,705 and US 2004/0138262, both of which are incorporated herein in the entirety. [0050] Normal blood glucose level: means a tested blood glucose level that is recognized as normal. For example, with the random blood sugar test, a random blood sugar level of 200 mg per deciliter (mg/dl) or higher suggests diabetes. With the fasting blood sugar test, the following ranges apply: less than 100 mg/dl-normal, from 100-125 mg/dl-prediabetic, and 126 mg/dl or higher on two separate tests-diabetic. With the oral glucose tolerance test (OGTT), the following ranges apply for the 120 minute blood glucose level: less than 140 mg/dl-normal, from 140-199 mg/dl-prediabetic, and over 200 mg/dl-diabetic. All of these tests are well known, standard and used routinely. [0051] Also in research use is the frequently sampled intravenous glucose tolerance test (FSIVGTT). [0052] HPA axis: means hypothalamic-pituitary-adrenal axis. [0053] Pseudo-Cushing's Syndrome: is a medical condition in which patients display some of the signs, symptoms and abnormal hormone levels seen in Cushing's Syndrome. Pseudo-Cushing's Syndrome is caused by a physiologic activation of the HPA axis and not by a neoplastic etiology that activates the HPA axis in Cushing's Syndrome or sub-clinical Cushing's Syndrome in which cortisol secretion is dysregulated. [0054] Pathologic activation of the HPA axis: This includes neoplastic conditions that produce excess cortisol levels. These conditions include Cushing's syndrome, which has a number of clinical features associated with the cortisol excess. Cushing's syndrome is caused by excess ACTH secretion from a corticotrope tumor or an ectopic ACTH-producing tumor, or from hyperplastic corticotrope cells under the stimulation of ectopic tumoral production of CRH; as well as primary adrenal disorders in which excessive amounts of cortisol secretion cause clinical Cushing's syndrome, such as primary pigmented adrenal nodular disease, macronodular adrenal hyperplasia, or other nodular disorders, for example, those seen in association with McCune Albright Syndrome. Autonomously functioning adrenal nodules, or other adrenal causes of Cushing's syndrome, may display dysregulated cortisol secretion that is insufficient to cause overt Cushing's syndrome; this has been termed "sub-clinical" Cushing's syndrome and may progress to clear-cut cortisol excess and Cushing's syndrome. [0055] Urine cortisol and other surrogate biochemical markers of HPA activation: This includes all tests currently available or subsequently developed that may be used to evaluate the effects of GRA dose and schedule on the HPA axis. Normal 24-hour urine free cortisol (UFC) levels range from 39-50 mcg/d when measured in most tandem mass spectrometry assays. Values are higher when measured by immunoassays because of cross-reactivity with the antibodies used in those assays with other steroids that are structurally similar to cortisol. Measurement of UFC by tandem mass spectrometry is the preferred method to assess the effect of GRAs on the HPA axis for three reasons: 1. ease of collection, 2. specificity of the assay (it measures only cortisol) and 3. the well-established range of values seen in physiologic stress, or Pseudo-Cushing's syndrome, which is 1-3 times the upper limit of the normal reference range. However, other biochemical tests may be used as surrogate biochemical markers of the HPA axis, including, but not limited to measurement of serum or salivary cortisol at multiple time-points throughout a 24 hour period, including both the day and night, which is expressed as either the mean value or the area under the curve. Normal ranges for all cortisol assays and 24-hour assessments will vary depending on the assay technique, and reagents, and must be used based on locally-determined values, or values provided by the manufacturer of an assay kit. The results obtained from any of these assays are then used to determine whether the dosage of the one or more GR antagonists need to be adjusted upwards or downwards so as to maintain 24-hour serum or urinary free cortisol levels at no more than about two or three times the upper level of the normal range, respectively. [0056] Insulin Insensitivity Disorder: means any disease or condition that is associated with insulin insensitivity, which includes, for example, diabetes, prediabetes, obesity, lipodystrophy,

hypertriglyceridemia, and/or growth hormone excess states. [0057] Assays for GR antagonists: as noted above, Clark, *Curr. Top. Med. Chem.*, 2008; 8 (9): 813-838 discloses in a review article known methods for evaluating GR antagonist properties by binding, functional and in vivo assays. However, high-throughput screening (HTS) is also used to rapidly screen many compounds for GR antagonist activity. Brown et al., *Biorganic & Medicinal Chemistry Letters*, Volume 21, Issue 1, 1 Jan. 2011, pages 137-140. Further, computational tools, including free available software, online services and commercial software is available for ADMET (Absorption, Distribution, Metabolism, Excretion and Toxicity) analysis. For example, the online service ADMET prediction offers assessment of human intestinal absorption, skin permeability and plasma protein binding. The online service PreADMET Toxicity Prediction predicts toxicological properties from chemical structures, including mutagenicity and carcinogenicity. See generally, *Nature*, Vol. 384, Supp. 7 Nov. 1996, Broach J et al. for a good discussion of high-throughput screening for drug discovery. See also Miner J. et al., *Molecular Endocrinology*, 17(1): 117-127 (2003) for the discovery of a non-steroidal GR antagonist using a high-throughput screen of a defined compound library using a GR-based competitive binding assay. The result was the identification of a non-steroidal GR antagonist characterized by a tri-aryl methane core structure exhibiting selective antagonist activity. Preferred assays are the luciferase reporter assays for nuclear receptors, including GR antagonists. See Paguin et al., *Current Chemical Genomics*, 2014, 4, 43-49. [0058] Non-rescue conditions: as used herein means treatment not conducted for the purpose of alleviating the side effects of a medication, or from cessation of a physiologic condition or from pathological causes of hypercortisolism, such as Cushing's Syndrome (see Pathologic activation of the HPA axis, above). As examples, the GR antagonists of the present invention are employed to improve insulin sensitivity, and not to provide rescue from the effects of anti-psychotic medications, such as weight gain, or to provide rescue from the effects of cessation of exercise and/or caloric restriction, such as adipose tissue mass gain, or to provide rescue from cancers, such as prostate cancer, in the form of treatment for those cancers. The term "non-rescue conditions" is also defined herein to exclude use of GR antagonists, including mifepristone, in patients with excessive or dysregulated cortisol secretion, such as those with Cushing's Syndrome or sub-clinical Cushing's Syndrome due to a functioning cortisol-producing adrenal adenoma. In contrast, the present invention involves the use of GR antagonists, such as mifepristone, to improve insulin sensitivity in people with normal cortisol regulation and under non-pathological conditions. However, the present invention explicitly includes treatment of Pseudo-Cushing's Syndrome, which can be diagnostically distinguished from Cushing's Syndrome. For example, the Dex-CRH test as well as a single measurement of cortisol in serum or saliva at bedtime has demonstrated high diagnostic accuracy in differentiating Pseudo-Cushing's Syndrome from Cushing's Syndrome. The Dex-CRH test is based on the fact that patients with Cushing's Syndrome are still able to mount a pituitary-adrenal response after administration of exogenous CRH despite pre-treatment with low-dose dexamethasone. Patients with Pseudo-Cushing's Syndrome, however, are relatively unresponsive to CRH stimulation when suppressed by dexamethasone. See Alwani et al. *European Journal of Endocrinology* (2014) 170, 477-486. [0059] Improve Insulin Sensitivity: as used herein means a reduction in insulin resistance and, thus, an improvement in how responsive human cells are to insulin. High insulin sensitivity allows the cells of the human body to use blood glucose more effectively to thus maintain a normal blood sugar. This improvement represents a reduced risk of many diseases, including type 2 diabetes. In contrast, the present invention does not seek to merely preserve insulin sensitivity, but rather to improve or alleviate insulin resistance. Consequently, the present invention may be readily distinguished from rescue methodologies that merely seek to prevent adipose rebound and insulin resistance without a need to improve impaired insulin sensitivity. See the definition of "non-rescue" conditions above. In contrast, the present invention improves insulin sensitivity under non-rescue conditions. For example, dysregulated cortisol secretion is an example of a rescue condition. The present invention improves insulin sensitivity in

people with normal cortisol regulation. [0060] Selective GR antagonism: means little or no activity, agonistic or antagonistic, for other receptors, such as androgen receptors. [0061] Pure GR antagonism: as used herein means a minimal level of GR agonism in vivo. [0062] GR antagonist daily dosage: means about 10 to 75 mg total of one or more GR antagonists per single dosage with more than one dose per day. Generally, two or more, and preferably three or more, dosages are administered daily, about every 4-8 hours. For example, a dosage regimen might entail administration of a 50 mg dosage every 6 hours with a total of 200 mg of one or more GR antagonists administered daily. Further, in normal people, higher doses of mifepristone can be associated with adverse events, such as rash and activation of the hypothalamic-pituitary-adrenal axis, hence a low dose-multiple administration per day regimen, such as a 50 mg dosage every 6 hours, is preferred. [0063] First Pass Effect: is the effect observed when mifepristone or another steroidal GR antagonist is taken orally. The mifepristone or other GR antagonist passes from the gut to the liver where the steroidal GR antagonist is metabolized. Thereby, the adipose tissue in the omentum and the liver "see" the highest concentration of the GR antagonist. Once past the hepatic metabolism, the systemic concentration of the steroidal GR antagonist is reduced and, thus, the hypothalamus and pituitary "see" these lower concentrations. Thus, a first pass effect is believed to be responsible for the reduced sensitivity of the dose-response curve of the hypothalamic-pituitary-adrenal axis compared to that for adipose tissue and the liver. However, the inventors are not bound by any theory in general, and do not intend for the invention disclosed herein to be so bound. Nonetheless, two important aspects of the invention disclosed herein are: 1) hepatic and adipose tissue insulin sensitivities are beneficially affected, and 2) HPA axis activation is minimized. These two important aspects of the disclosed invention are accentuated by use of low dose-multiple daily administration of one or more GR antagonists with an upper limit on total daily dosage.

Consideration of the Mode of Administration, Dose and Schedule of Administration of GR Antagonists

[0064] The present inventors have discovered that administration of mifepristone or one or more metabolites thereof, either orally, by depot or transdermally (by patch, for example) to patients in specific dosages or dosage ranges and/or times of the day might reverse elements of the metabolic syndrome.

[0065] Furthermore, the inventors considered that activation of the hypothalamic-pituitary-adrenal axis might override the effects of glucocorticoid antagonism. This would occur at antiglucocorticoid doses or schedules of administration that antagonize cortisol's inhibitory effect on ACTH secretion, leading to an increase in ACTH and cortisol, and potentially diminishing or abolishing an effect on insulin insensitivity should systemic levels of GR antagonist not be sufficient to block the effects of increased cortisol. Thus, the present inventors considered that the dosing regimens should be selected that differentially block cortisol action at peripheral levels, but which do not activate the hypothalamic-pituitary-adrenal axis to an extent that would override the peripheral blockade.

[0066] The present inventors have, in fact, discovered that glucocorticoid antagonists, such as mifepristone, metapristone or a derivative, analog or prodrug of these compounds may be used advantageously to improve insulin sensitivity. The present inventors have further discovered that one or more GR antagonists, generally, may be used to improve insulin sensitivity whether these GR antagonists are steroidal or non-steroidal.

[0067] Generally, dosages of one or more glucocorticoid antagonists are administered in the amount of from about 500 micrograms (0.5 mg) to about 100 mg once or more per day. For example, 50 mg of mifepristone may be administered every six (6) hours to a patient having abnormal glucose tolerance, thereby improving insulin sensitivity. The treatment regime is adjusted until the patient's blood glucose level attains and maintains a normal level as measured by a standard known glucose test. A treating physician may, of course, reduce or increase dosages as deemed advisable and routinely monitor the effectiveness of dosages.

[0068] Generally, the same dosage range may be used for any one or more GR antagonists when used in accordance with the present invention to increase insulin sensitivity.

Testing for Cortisol Levels to Evaluate the Need for Adjustment of Dosage of One or More GR Antagonists for Sequent Administration of the One or More GR Antagonists

[0069] In accordance with another aspect of the present invention, insulin, glucose and 24-hour serum or urine cortisol levels are measured once steady-state blood levels are reached, at five or more half-times of the clearance of one or more GR antagonists to determine a more preferred dosage of the one or more GR antagonists for sequent administration. It is considered that if the dosage of the one or more GR antagonists is too low, insulin sensitivity will not be improved, whereas if the dosage thereof is too high the hypothalamus-pituitary-adrenal (HPA) axis will be activated, and too much cortisol will be produced. To achieve maximum dose efficiency, insulin, glucose and cortisol testing should be conducted prior to dose adjustments for sequent administration of the one or more GR antagonists to improve insulin sensitivity without activating the HPA.

[0070] The GR antagonists employed in the present invention may be any of those disclosed in U.S. Pat. Nos. 5,959,058; 4,296,206; 4,386,085; 4,447,424; 4,477,445; 4,519,946' 4,540,686; 4,547,493; 4,634,695; 4,634,696; 4,753,932; 4,774,236; 4,808,710, 4,814,327; 4,829,060; 4,861,763; 4,912,097; for example. Each and all of these patents are incorporated by reference herein in the entirety. However, preliminary assays of compounds exhibiting GR antagonism will ensure that only the GR antagonist compounds having the most selective and purest GR antagonism will be used.

[0071] Examples of steroidal GR antagonists that may be used in the present invention are, for example, cortexolone, dexamethasone-oxetanone, 19-nordeoxycorticosterone, 19-norprogesterone, and cortisol-21-mesylate.

[0072] While many GR antagonists are steroidal compounds, non-steroidal GR antagonists may also be used in the present invention. For example, mention may be made of the tri-aryl methane compounds, such as [bis(4-N,N-dimethylaminophenyl)(2-chloro-5-nitrophenyl)methane]. This compound, for example, is a GR-specific antagonist that binds with nanomolar affinity to the GR while having no detectable affinity for the related receptors for mineralocorticoids, androgens, estrogens and progestins. See Miner, cited above. Other tri-aryl methane compounds may be ascertained using the well-known competitive binding assays disclosed in Clark and Miner, both cited above. Further, use may also be made of high-throughput screening as disclosed in Broach & Thomer, cited above.

[0073] In particular, the known compounds ORIC-101, Relacorilant and/or Miricorilant are preferred GR antagonists for use in the present invention. The molecular structures of ORC-101 is shown in appendix structure B2. The molecular structure of Relacorilant is shown in appendix structure C, and that of Miricorilant in appendix structure D. The molecular structure of Exicorilant, which may also be used, is shown in appendix structure E.

[0074] Generally, whether steroidal or non-steroidal GR antagonists are used, the applicable dosages range from 10 to 75 mg in total per dose with doses being administered from 1 to 6 times per day. The treating physician may adjust the number of doses per day as deemed required based upon measured serum, glucose, insulin and cortisol levels.

[0075] It is preferred, however, to use dosages of from about 30 to 70 mg per dose of one of more GR antagonists with administration of 3 to 5 times daily. For example, 50 mg of mifepristone can be administered every 6 hours with 200 mg being the total daily dose. Generally, the total daily dose of one or more GR antagonists will not exceed about 500 mg daily in total.

[0076] Although any mode of administration may be used for the one or more GR antagonists, such as oral, intravenous and/or intramuscular injection or topical, oral administration is preferred.

[0077] The following examples are provided only for purposes of illustration and are not intended to be limitative.

Example 1

[0078] A randomized, placebo-controlled, double-blind crossover trial was conducted to evaluate the use of an anti-glucocorticoid as a means for ameliorating insulin resistance. Mifepristone, 50 mg, or a placebo, was administered every six hours for nine days to sixteen overweight or obese (BMI 25-37 kg/m²) subjects (seven women). Participant ages ranged from 48 to 64 years. Ten had prediabetes and six had mild diabetes type II (HgbA1c less than 7%). Treatment of the glucose disorder with diet or a stable dose of metformin (used by two patients) was maintained for three (3) months before the study. The interventions were nine days of mifepristone or placebo treatment followed by a six (6) to eight (8) week wash out and cross over to the other arm. At baseline and at the end of each treatment period, parameters of insulin sensitivity were assessed by a frequently sampled intravenous glucose tolerance test (FSIVGTT) and an oral glucose tolerance test (OGTT). A whole body glucose disposal index of insulin sensitivity (S_{sub.I}) was derived using a minimal model analysis. Adipose tissue-S_{sub.I}, which reflects insulin-induced free fatty acid suppression, was estimated by the log-linear slope of free fatty acid levels and the AUC of insulin during the FSIVGTT. Matsuda index, oral glucose insulin sensitivity index (OGIS, ml/min/m²), and hepatic insulin resistance index (HIRI, mg/dl×microU/ml×10⁵) were derived from OGTT. Adipo-IR, a surrogate measure of adipocyte insulin resistance, was calculated by multiplying fasting concentrations of free fatty acids and insulin (mmol×pmol/L). Comparison of various post-treatment parameters was performed using crossover ANOVA. This analysis takes into account specific treatment arm, treatment order, and treatment effects. Thus, carryover effects were explicitly tested for in the crossover study.

[0079] The glucocorticoid receptor blockade did not affect body weight (p=0.59). Fifteen subjects had urine free cortisol values less than three-times normal, and eight of these were within the normal range. One subject had elevated values at all time-points before and during both mifepristone and placebo. Mifepristone administration for one week did not alter whole body glucose disposal indices of insulin sensitivity (FSIVGTT-S_{sub.I}: p=0.60, Matsuda Index: p=0.16, and OGIS: p=0.92).

[0080] However, glucocorticoid receptor blockade significantly decreased fasting insulin levels when compared with placebo (mean±SD; 95.7±76.1 vs 142.8±102.2pmol/L, p =0.0006), as well as fasting glucose levels (mean±SD, 100.4±15.3 vs 107.8±15.7 mg/dL). Glucocorticoid receptor blockade also reduced HIRI (70±44.3 vs 50.2±38.7, p=0.08) and Adipo-IR (65.5±43.8 vs 50.0±46.0, p<0.001) and improved Adipose SI (42.8±23.9 vs 61.7±32.9, p<0.001).

[0081] Thus, in patients with abnormal glucose tolerance, short term glucocorticoid receptor blockade improves hepatic and adipose tissue insulin sensitivity without significantly altering whole body sensitivity. This result also is important for the additional reason that insulin resistance in individual tissues can cause reduced skeletal muscle metabolic flexibility.

[0082] Examples 1 and 2 are from the same clinical protocol, however example 2 provides additional data and analysis thereof that became available after the initial data of Example 1 were processed and analyzed.

Example 2

[0083] To further demonstrate the effectiveness of the present invention in improving insulin sensitivity under non-rescue conditions with patients having normal cortisol dynamics, the following randomized, triple-blinded, placebo-controlled, cross-over study was conducted using mifepristone as an exemplary and representative GR antagonist.

[0084] The inventors conducted a random, triple-blinded, placebo-controlled, cross-over study using mifepristone (50 mg every 6 hours, 200 mg total daily dose) in overweight/obese individuals (n=16, 44% female) with pre-DM (pre-diabetes mellitus) or mild type 2 diabetes mellitus (glycated hemoglobin A1C greater or equal to 7). Mifepristone or placebo was administered for 9 days followed by a washout period of 6 to 8 weeks, and then crossover to the other treatment arm. At baseline and after each treatment period, the oral glucose tolerance test and frequently sampled

intravenous glucose tolerance test were performed. Insulin sensitivity (SI) was measured using Matsuda index and oral glucose insulin sensitivity index (OGISI). Hepatic S.sub.I was assessed using hepatic insulin resistance index (HIRI), and adipose tissue S.sub.I was assessed using adipose-IR adipose tissue insulin sensitivity index (ATISI).

Methods

Trial Design

[0085] The study in this example used a prospective, randomized, double-blind placebo-controlled, cross-over design in which eligible subjects were randomized in blocks of six to the initial arm of the study: either placebo or mifepristone (50 mg tablets every 6 hours) for 9 days. This was followed by a 6-8 week washout period. Subjects were then crossed over to the other treatment arm for an additional 9 days. The study was conducted at the National Institutes of Health (NIH) Clinical Center, Bethesda, Maryland, U.S.

Participants

[0086] Subjects eligible for this study included men and women 35-70 years of age who were overweight or obese (BMI 25-37 kg/m^{sup.2}) with pre-diabetes (defined as fasting glucose of 100 to 125 mg/dl or a 2-hour postprandial glucose value of greater than or equal to 140 mg/dl and less than 200 mg/dl during an oral glucose tolerance test (OGTT) or mild diabetes (defined as a HgbA1C greater than 6.5% to less than or equal to 7%). Eligible subjects took no medications or were on a stable dose of metformin and no other hypoglycemic agents for at least 3 months before beginning the study. The study excluded pregnant and lactating women and subjects with uncontrolled hypertension (>180/110 mmHg) or any unstable medical conditions (e.g., impaired cardiac function, severe respiratory insufficiency), liver function tests (ALT and AST) more than 3 times the normal upper limit, severe renal impairment (creatinine clearance<30 ml/min), evidence of HIV or hepatitis C infection, history of hemorrhagic disorders or use of anticoagulants, history of endometrial cancer or hyperplasia, unexplained vaginal bleeding or endometrial thickness greater than 6 mm, changes in lipid lowering medication dose within 1 month of study entry, use of any medications or dietary supplements with strong inhibition of CYP3A4, use of grapefruit juice or herbal supplements within 14 days of study drug initiation, use of glucocorticoids or megestrol within 6 months of study initiation, use of estrogen-based hormone therapy, depression (evaluated by PHQ-9 with a cut off of greater or equal to 10), excessive alcohol use (>2 drinks per day for women and >3 drinks per day for men), untreated thyroid dysfunction, moderate/severe anemia (hemoglobin<10 g/dL), blood donations totaling 500 mL or more within one month of study initiation, prolonged QTc interval, and subjects in a weight loss program or who were actively dieting. Perimenopausal and postmenopausal women underwent a transvaginal ultrasound to evaluate endometrial thickness, and a pelvic MRI was performed if the transvaginal ultrasound did not provide an accurate reading of the endometrial thickness. See FIG. 3, which provides a flow diagram of participants throughout the study. Subjects were required to be weight-stable for the 2 weeks prior to entry and not to start any new diet/exercise programs during the study.

[0087] Study subjects were screened and evaluated at the NIH Clinical Center for eligibility. 63 subjects were consented and screened for eligibility for the study protocol, with 44 subjects not studied based on inclusion and exclusion criteria. The remaining 19 subjects were initially enrolled in the study, with a final sample of 16 subjects completing the study with sufficient data for analysis.

Study Approval

[0088] The study was approved by the Institutional Review Board of the National Institute of Child Health and Human Development, and all procedures were performed as per the institutional guidelines. Eligible participants provided written informed consent to participate in protocol 11-CH-0208 (Clinicaltrials.gov; NCT01419535).

Sample Size

[0089] Patient recruitment and assessment for eligibility is shown in FIG. 3. Initially, 19 subjects

underwent randomization. One subject dropped out, while 2 other subjects had malfunctioning intravenous access resulting in incomplete data. The final analysis was performed on 16 subjects. Though the protocol listed a sample size goal of 17 subjects, analysis for 16 subjects was used. Beyond this, there were no changes to the protocol methodology after trial commencement. An interim analysis was conducted to evaluate the effect of the mifepristone dose (50 mg/6 hr) on the hypothalamic-pituitary-adrenal axis. The study statistician was unblinded to the first 6 subjects (one randomization block) to evaluate whether the 24-hour UFC levels following mifepristone treatment were equal to, or more than twice, the upper limit of the reference range in 4 or more of the 6 subjects. In that scenario, the mifepristone dose would be decreased to 25 mg every 6 hours (100 mg/day). However, UFC values remained below the threshold and no dose adjustment was required.

Randomization, Blinding and Study Drug Management

[0090] Mifepristone was obtained through an active IND under a CRADA agreement partnership with Laboratoire HRA-Pharma (Paris, France). Mifepristone and inert, look-alike placebo were packaged into 50 mg capsules by the NIH Clinical Center Pharmaceutical Development Service.

Procedures of the Study

[0091] To reduce potential variability associated with estrogen in premenopausal women, testing was performed in the early follicular phase (days 1-7 of menstrual cycle). Baseline height and weight were measured after an overnight fast. Subjects received an isocaloric regular diet during the inpatient part of the study, and those on metformin took the same dose throughout the study. Baseline testing took place over three inpatient days at the NIH Clinical Center, during which fasting blood was drawn, 24-hour urine was collected, 24-hour serial blood draws were obtained, and patients received both a frequently sampled glucose tolerance test (FSIVGTT) and a 75-g oral glucose tolerance test (OGTT). Subjects were then randomized to receive placebo or mifepristone for 9 days, of which the first 6 days the subjects returned home, and the final 3 days subjects underwent the same metabolic testing as during the baseline period. Subjects randomized to the mifepristone treatment arm were instructed to take one capsule (50 mg) every 6 hours (200 mg/day) over the course of 9 days. The placebo treatment arm received the same instructions to take one capsule every 6 hours over the course of the 9-day treatment period. Following the completion of the post-drug testing, subjects entered a 6-8 week washout period, after which they were crossed over to the other treatment arm, serving as their own controls for the study drug treatments.

[0092] All subjects were instructed to record any adverse effects and to report them to study personnel, and women were advised to keep a record of any abnormal menstrual bleeding. For proper safety monitoring, subjects returned approximately one week (day 19) and three weeks (day 33) after the discontinuation of the study drug for both treatment arms. During this safety visit, serum electrolytes, renal and liver function, and complete blood count were measured, as well as evaluation and treatment for potential treatment-related abnormalities/adverse events.

Study Interventions

Frequently Sampled Intravenous Glucose Tolerance Test

[0093] Insulin sensitivity index, SI, was determined from the minimal model using data from the insulin-modified FSIVGTT as previously described (version 6.02; MinMOD Millennium, Los Angeles, CA). Briefly, after an overnight fast, on day 2 of baseline and post-treatment testing, two intravenous catheters were placed, one in each arm or in the antecubital veins. A 50% dextrose (0.3 gm/kg) bolus was infused intravenously over 1 minute at timepoint 0-min, and an insulin (0.03 units/kg) bolus was infused at the 20-min timepoint. Several blood draws were obtained at: -10, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 19, 22, 23, 24, 25, 27, 30, 40, 50, 60, 70, 80, 90, 100, 120, 150 and 180 minutes for measurement of glucose, insulin and free fatty acids.

Oral Glucose Tolerance Test

[0094] On day 3 of baseline and post-treatment testing, subjects underwent a 75-gram OGTT.

Serial blood draws for measurement of glucose, insulin and c-peptide were obtained at -5, 0, 30, 60, 90, 120, 150, and 180 minutes.

Study Endpoints

Surrogate Measures of Insulin Sensitivity and β -Cell Function

[0095] The Matsuda index and Oral Glucose Insulin Sensitivity Index (OGIS), surrogate indices of whole-body insulin sensitivity were derived from plasma glucose and insulin concentrations during the OGTT as previously described. Abdul-Ghani MA, et. al., *Diabetes Care* 2007; 30 (1): 89-94. HIRI was calculated as the product of the area under the curves of glucose and insulin for the first 30 minutes of the OGTT. Abdul-Ghani MA, *ibid.* The Adipo-IR index, a surrogate measure for fasting adipose tissue insulin resistance, was calculated as the product of fasting insulin and fasting FFA. Sondergaard E. et al., *J. Clin. Endocrinol. Metab.* 2017; 102(4): 1193-9. A dynamic surrogate index of adipose tissue insulin sensitivity was assessed by calculating insulin's ability to suppress FFA levels during a FSIVGTT. Carpenter A. et al., *Diabetes* 2002; 51(6):1867-75 and Trottier A. et al. *Fertil. Steril.* 1992; 98 (6): 1627-34. The adipose tissue insulin sensitivity index was calculated as the ratio of the slope of the linear decrease in natural log transformed FFA [Ln (FFA) slope] during the first 90 minutes of the FSIVGTT and the AUC of insulin during that 90-minute period (AUC Insulin.sub.0-90 min). Mathematical modeling of C-peptide concentrations was used to estimate insulin secretory rates and calculate β -cell responsiveness parameters (basal ($\emptyset B$); first-phase ($\emptyset 1$); second phase ($\emptyset 2$); and total ($\emptyset TOT$) as previously described. Toffolo G. et al., *Metabolism* 1999; 48 (9): 1162-6 and Cobelli C. et al. *Am. J. Physiol. Endocrinol. Metab.* 2007; 293 (1): E1-E5. Assuming an absence of hepatic C-peptide extraction, postprandial insulin clearance was calculated as the ratio of mean insulin secretion and insulin concentration. Trico D. et al. *JCI Insight* 2018; 3(24):e124912.

Statistical Analysis

[0096] Changes in insulin sensitivity as measured by FSIVGTT were prospectively designated as the primary endpoint for this study. All other comparisons were considered secondary. Continuous variables were described as mean and standard deviation and categorical data were represented as proportions and frequencies. All continuous data were examined for normality and homogeneity of variances and were either log transformed as appropriate or assessed using non-parametric tests. Between and within treatment comparisons were performed using mixed models for a crossover design that also considers sequence effect, if any. Treatment-by-period interaction was tested for primary outcome. Two-sided p-values were used for all analyses. Repeated measures were analyzed by repeated-measure ANOVA including treatment group time, and treatment-time interaction as factors. Post-hoc pairwise comparisons were performed by Bonferroni's test. The level of statistical significance was set a priori at <0.05 and the p-values were not adjusted for multiple comparisons. Data were analyzed with JMP version 13.0 (SAS Institute, Cary, NC) and GraphPad Prism 7 (GraphPad Software Inc., La Jolla, CA).

Results

[0097] There were no significant differences observed in total body insulin sensitivity after mifepristone treatment. However, the AUC for insulin was significantly lower after mifepristone treatment compared to placebo. Mifepristone treatment resulted in significant improvement in adipose tissue sensitivity, and there was a near-significant improvement in hepatic sensitivity. The conclusion is that short-term GR antagonist use, as exemplified by mifepristone, improves adipose and hepatic insulin sensitivity among overweight or obese individuals with hyperglycemia without inducing hypercortisolism.

Baseline Characteristics of Study Subjects

[0098] The baseline characteristics of the 16 subjects who completed the study are described in table 1 below.

TABLE-US-00001 TABLE 1 Clinical and Metabolic Characteristics of Subjects at Baseline: n = 16
Age (years) 55.5 ± 7.9 Sex (F/M) 7/9 Weight (kg) 97.1 ± 15.6 Body-Mass Index (kg/m³) $32.2 \pm$

3.7 Systolic Blood Pressure (mmHg) 121 ± 17 Diastolic Blood Pressure (mmHg) 69 ± 10 Total Cholesterol (mg/dL) 181.1 ± 52.4 HDL Cholesterol (mg/dL) 44.7 ± 9.6 LDL Cholesterol (mg/dL) 107.8 ± 39.3 Triglycerides (mg/dL) 142.6 ± 83.8 Free Fatty Acids (mmol/L) 0.613 ± 0.195 Fasting Glucose (mg/dL) 102.7 ± 11.6 Fasting Insulin (pmol/L) 138.9 ± 89.0 HbA1C (%) 5.98 ± 0.60 HOMA-IR 5.16 ± 3.49 QUICKI 0.310 ± 0.025 OGIS (ml/min/m.sup.2) 283.9 ± 66.3 Matsuda Index 2.15 ± 1.05 S.sub.I (min.sup.-1 .Math. μ U .Math. ml.sup.-1) 1.06 ± 0.61 Urinary Free Cortisol (μ g/24 hrs) 22.9 ± 17.2

[0099] For the above listed characteristics, the following is noted. 44% of the participants were female. Participants were overweight or obese with a mean BMI of 32.2 kg/m.sup.2 (range: 25.7 to 37.5 kg/m.sup.2). 63% of the cohort (n=10) were prediabetic [impaired fasting glucose (IFG), 13% n=2; impaired glucose tolerance (IGT), 31%, n=5; and both IFG and IGT, 19%], n=3, and 37%, n=6 were diagnosed with T2DM. The average (HgbA1C) ($\pm$ SD) at baseline of the entire, prediabetic, and diabetic groups was $5.98 \pm 0.57\%$, $5.88 \pm 0.43\%$ and $6.26 \pm 0.59\%$, respectively. Two of the diabetic patients were taking stable doses of metformin. The mean systolic and diastolic blood pressures (BP), were 121 ± 17 and 69 ± 10 mm of Hg, respectively. 63% of the participants (n=10) were on anti-hypertensive agents and 31% (n=5) were on statins. 38% of the participants had low levels of HDL cholesterol (<40 mg/dl (men) or <50 mg/dl (women)) or high triglyceride (TG) levels (>150 mg/dL). Surrogate measures of insulin sensitivity indices based on glucose and insulin levels during fasting (quantitative insulin sensitivity check index, QUICKI, and index of homeostasis model assessment, HOMA-IR) and glucose tolerance tests (oral glucose insulin sensitivity index, OGIS and Matsuda index from oral glucose tolerance test (OGTT) and insulin sensitivity index derived from the frequently sampled intravenous glucose tolerance test (FSIVGTT), S.sub.I) are reported in Table 1. HOMA-IR>2.5, QUICKI less than or equal to 0.331, OGIS less than or equal to 344, Matsuda Index less than or equal to 2.5, or S.sub.I less than or equal to 2.1 has been used to define insulin resistance. Owei I. et al. *BMJ Open Diabetes Res. Care* 2017; 5 (1): e000415; Kernan WN et al. *Stroke* 2013; 34(6): 1431-6; Antoniolli LP et al. *Diabetol. Metab. Syndr.* 2018; 10:65; Ascaso JF et al. *Diabetes Care* 2003; 26(12): 3320-5. Based on the mean values of HOMA-IR, QUICKI, OGIS, Matsuda index and .sub.SI, subjects in this cohort were insulin resistant. See Table 1. The mean 24-hour urinary free cortisol (UFC) values were within normal limits (22.9 ± 17.2 μ g/24 hr.; normal: 3.5-45 μ g/24 hr.) The final study population exhibited the inclusion criteria set for the trial.

Effects of Mifepristone Treatment on Circulating Levels of Cortisol and ACTH and 24-hour Urinary Free Cortisol

[0100] Mifepristone is known to block the negative feedback effects of cortisol, which leads to elevations in ACTH and cortisol, by antagonizing GRs. Gaillard RC et al., *Proc Natl Acad Sci U S A.* 1984; 81 (12): 3879-82. Mean ACTH and cortisol levels were higher during mifepristone than placebo administration specifically during mornings. See FIG. 4. ACTH levels were significantly higher from 5 AM-10 AM, while the cortisol difference also started around 5 AM, the levels remained significantly elevated until about 5 PM when compared with placebo. See FIG. 4. The area under the curve (AUC) measurements for ACTH and cortisol were significantly higher during mifepristone treatment (574.5 ± 260.8 vs. 448.7 ± 193.5 pg/mL.Math.min for ACTH, $p < 0.001$, and 255.4 ± 61.9 vs. 151.1 ± 21.1 μ g/dL.Math.min for cortisol; $p < 0.001$). Overall, mifepristone significantly increased 24-hour UFC compared with placebo (74.1 ± 75.3 vs, 25.3 ± 30.2 μ g/24 hr. $p = 0.0001$). These stimulatory effects of mifepristone were not uniform, with post-treatment mean UFC above the normal limit in 8 subjects (120.4 ± 83.9 μ g/24 hr). In the mifepristone treatment arm, serum cortisol AUC levels were significantly higher in the subjects with higher 24-hr UFC compared to the subjects with normal 24-hr UFC (284.9 ± 39.9 vs, 229.8 ± 68.5 μ g/dL.Math.min for cortisol $p = 0.009$. Plasma DHEA sulfate (DHEAS) and androstenedione levels were significantly higher after mifepristone therapy when compared with placebo. See Table 2 below. Overall, these results indicate that an exemplary dosage of mifepristone (50 mg tablets every 6 hours) for 9 days

substantially increased cortisol and ACTH levels as a compensatory response to GR blockade.

TABLE-US-00002 TABLE 2 Clinical and Metabolic Parameters and Hormone Levels After Placebo and Mifepristone Administration: P Placebo Mifepristone value Weight (kg) 97.7 ± 16.3 97.4 ± 8 0.59 Body-Mass Index (kg/m^{sup.2}) 32.4 ± 4.3 32.2 ± 3.9 0.39 Systolic Blood 121 ± 12 130 ± 13 0.009 Pressure (mmHg) Diastolic Blood 71 ± 9.5 77 ± 9.2 0.01 Pressure (mmHg) Total Cholesterol (mg/dL) 177.3 ± 38.7 163.6 ± 41.9 0.004 HDL Cholesterol (mg/dL) 45 ± 8.8 41.3 ± 11.9 0.01 LDL Cholesterol (mg/dL) 101.9 ± 31.1 97.1 ± 34.6 0.10 Triglycerides (mg/dL) 152 ± 79.3 125.6 ± 65.6 0.002 Total testosterone 355.2 ± 124.1 235.2 ± 87.5 0.01 (male) (ng/dL) Androstenedione (ng/dL) 66.7 ± 41.7 110 ± 64.7 0.007 DHEA Sulfate (μ g/mL) 0.82 ± 0.45 1.04 ± 0.64 0.004 Urinary Free 25.3 ± 30.2 74.1 ± 75.3 0.0001 Cortisol (μ g/24 hr)

[0101] Post-treatment values shown are unadjusted means $\pm$ SD. Post-treatment values were adjusted for baseline value and treatment group using mixed-model regression. P values indicate significance for comparisons between mifepristone and placebo.

[0102] FIG. 4 shows a serial time course of plasma ACTH (A) and cortisol (C) every hour from 9 AM one day to 9 AM the following day. Data shown are mean $\pm$ SEM with p values provided for the effects of treatment group, time, and the treatment-time interaction, which was obtained with repeated-measures ANOVA with post-hoc Bonferroni's test. Tukey box and whisker plots show ACTH (B) and cortisol (D) area under the curve for the 24-hour period. Top and bottom box limits represent the 75^{sup.th} and 25^{sup.th} percentile, respectively; midlines represent the median, and top and bottom whiskers represent the 75^{sup.th} percentile plus 1.5 of the interquartile range (IQR) and the 25^{sup.th} percentile minus 1.5 IQR, respectively. Post-treatment values were adjusted for baseline and treatment group using mixed-model regression. P values indicate significance for comparisons between mifepristone and placebo.

Effect of Mifepristone Treatment on Body Weight, Blood Pressure (BP) and Serum Lipids

[0103] Changes in body weight, BP and serum lipid concentrations for each study subject were evaluated at baseline and after 9-day administration of mifepristone or placebo. There were no significant differences observed in body weight or BMI between the treatment arms. See Table 2. BP was higher after mifepristone treatment for both systolic and diastolic values. Mifepristone treatment was associated with significantly lower serum total cholesterol, HDL cholesterol and triglycerides. See Table 2. The mean plasma LDL levels were lower after mifepristone treatment, but the difference was not statistically significant.

Effect of Mifepristone on Plasma Glucose, Serum Insulin and Indices of Insulin Sensitivity

[0104] Mifepristone treatment resulted in significant lowering of plasma glucose and insulin. See Table 3 below.

TABLE-US-00003 TABLE 3 Effects of Placebo and Mifepristone Administration on Metabolic Parameters and Measures of Insulin Sensitivity and Resistance P Placebo Mifepristone value Fasting Glucose (mg/dL) 107.8 ± 15.7 100.4 ± 15.3 <0.001 Fasting Insulin (pmol/L) 142.8 ± 102.2 95.6 ± 76.1 <0.001 HOMA-IR 5.78 ± 4.92 3.58 ± 3.27 <0.001 QUICKI 0.310 ± 0.029 0.332 ± 0.034 <0.001 S.sub.I (min^{sup.-1} .Math. μ U .Math. ml^{sup.-1}) 1.41 ± 0.87 1.49 ± 1.17 0.60 OGIS (ml/min/m^{sup.2}) 308.9 ± 72.9 305 ± 48.9 0.92 Matsuda Index 2.52 ± 1.96 2.67 ± 1.03 0.16 Hepatic Insulin Resistance 70 ± 44.3 50.2 ± 38.7 0.08 Index (HIRI) Adipose-tissue Insulin 65.5 ± 43.8 50.0 ± 46.0 <0.001 Resistance Index (Adipo-IR) Adipose-tissue Insulin 42.8 ± 23.9 61.7 ± 32.9 <0.001 Sensitivity Index

[0105] Post-treatment values shown are unadjusted means $\pm$ SD. QUICKI, Quantitative Insulin Sensitivity Check Index; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; OGIS, Oral Glucose Insulin Sensitivity; S.sub.I, Insulin Sensitivity Index derived from the insulin-modified frequently sampled intravenous glucose tolerance test. Post-treatment values were adjusted for baseline value and treatment group using mixed-model regression. P values indicate significance for comparisons between mifepristone and placebo.

[0106] The results shown in Table 3 indicate that mifepristone significantly reduced HOMA-IR and improved QUICKI as compared with placebo. The primary endpoint, insulin sensitivity index (S.sub.I) as measured by FSIVGTT was not significantly different between the treatment arms. See Table 3 and FIG. 6A. The time course of glucose and insulin concentrations during the OGTT were not significantly different between the treatment groups. See FIGS. 6A and 6C. The AUC measurements for glucose during OGTT appeared to be higher in the mifepristone group, but this apparent difference was not statistically significant. See FIG. 6B. However, AUC for insulin was significantly lower by about 25% during mifepristone treatment as compared to placebo. See FIG. 6D. Overall glucose insensitivity indices derived from OGTT, OGIS and Matsuda index were not significantly different between placebo and mifepristone. See Table 3 and FIGS. 6B and 6C. Hepatic insulin resistance index (HIRI) tended to be lower following mifepristone administration ($p=0.08$). See Table 3 and FIG. 6D. Adipose tissue insulin resistance was assessed during the fasting state (Adipo-IR) and a measure of insulin-induced suppression of free fatty acids (FFA) levels during FSIVGT adipose tissue insulin sensitivity index (Adipo-SI) was estimated by the log-linear slope of FFA levels during FSIVGTT corrected for circulating insulin measured (AUC) during the same period. Mifepristone administration was associated with a significant attenuation in Adipo-IR and increase in Adipo-SI. See FIGS. 6E and 6F.

[0107] FIG. 5 shows the results of a time course of plasma glucose (A) and insulin (C) during OGTT. Data shown are mean $\pm$ SEM. Values for the effects of treatment group, time and the treatment-time interaction were obtained with repeated-measures ANOVA with post-hoc Bonferroni's test. Tukey box and whisker plots of glucose (B) and insulin (D) area under the curve for the duration of 3-hr OGTT. Top and bottom box limits represent the 75.sup.th and 25.sup.th percentile, respectively; midlines represent the median; and top and bottom whiskers represent the 75.sup.th percentile plus 1.5 of the interquartile range (IQR) and the 25.sup.th percentile minus 1.5 IQR, respectively. Symbols represent values that lie outside the range of the whiskers. Post-treatment values were adjusted for baseline value and treatment group using mixed-model regression. P values indicate significance for comparisons between mifepristone and placebo.

[0108] FIG. 6 shows the effects of placebo and mifepristone on indices of insulin sensitivity. Tukey box and whisker plots (A) S.sub.I, insulin sensitivity index derived from the insulin-modified frequently sampled intravenous glucose tolerance test, (B) Matsuda index, (C) OGIS, oral glucose insulin sensitivity index, (D) hepatic insulin resistance index (HIRI), (E) adipose tissue insulin resistance index, and (F) adipose tissue insulin sensitivity index. Top and bottom box limits represent the 75.sup.th and 25.sup.th percentile, respectively; midlines represent the median; and top and bottom whiskers represent the 75.sup.th percentile plus 1.5 of the interquartile range (IQR) and the 25.sup.th percentile minus 1.5 IQR, respectively. Symbols represent values that lie outside the range of the whiskers. Post-treatment values were adjusted for baseline value and treatment group using mixed-model regression. P values indicate significance for comparisons between mifepristone and placebo.

Effect of Mifepristone on Insulin Secretory Rates, Insulin Clearance and β -Cell Responsivity Indices

[0109] Insulin secretory rates (ISR) were derived by mathematical modeling of C-peptide and glucose concentrations during an OGTT. β -cell sensitivity parameters derived from the modeling include basal ($\emptyset$.sub.B), first-phase ($\emptyset$.sub.1), second-phase ($\emptyset$.sub.2) and total ($\emptyset$.sub.TOT). Fasting ISR was significantly lower following mifepristone (see FIG. 7B), but the ISR during OGTT was not different. See FIG. 7A. The static, dynamic and total β -cell responsivity indices derived from OGTT were similar in the mifepristone and placebo groups. See FIG. 8. Insulin clearance during OGTT was lower following mifepristone when compared with placebo. See FIG. 7B.

No Serious Adverse Effects Associated With Mifepristone and Placebo Administration

[0110] No serious adverse events were reported by subjects or detected independently during the

experiments in this Example. The safety laboratory results remained within normal limits except for two subjects with increased alanine transferase after mifepristone and one patient before and after, and another after placebo administration. Abnormal values were less than one-fold the upper limit of normal 40 U/L, ranging from 41 to 47 U/L.

[0111] Example 2 demonstrates the effects of mifepristone on insulin sensitivity in overweight and obese individuals who had no secondary reason for abnormal glucose control, such as an adrenal adenoma or diabetogenic medication. This point reinforces the fact that the use of one or more GR antagonists in accordance with the present invention to improve insulin sensitivity is effected under non-rescue conditions, i.e., secondary reasons for abnormal glucose control other than being overweight or obese.

[0112] Specifically, the present inventors have demonstrated for the first time in such a population (overweight and/or obese) that short term GR antagonism had variable effects on insulin sensitivity, with a significant improvement at the adipose tissue, a tendency to such improvement at the liver, and no significant difference in overall peripheral insulin sensitivity. These salutary changes were accompanied by significant decreases in fasting glucose and insulin levels and the insulin area under the curve (AUC) values during an OGTT. The improvements in glycemic control can be attributed to mifepristone, as weight did not change. We surmise that the increased adipocyte sensitivity, with decreased lipolysis (as shown by decreased FFA excursions), led to partial improvement in hepatic sensitivity, and that with longer exposure to one or more GR antagonists, whole body insulin sensitivity would improve. However, the present invention explicitly contemplates improving insulin sensitivity in individuals who are neither overweight nor obese. It has been estimated that about 12% of people in the U.S. with diabetes are normal weight. Such people may have as much fat as someone who is overweight, but the fat of such normal weight individuals is visceral fat.

[0113] The study in Example 2 demonstrates many important features of the present invention. Further, the cross-over experimental design reduced the potential for inter-person variability and increased the ability to detect treatment differences in the modest sample size (n=16). Also, the use of a wash-out period of 6-8 weeks provided a minimization of carryover effects of treatment. As noted, efforts were made to maintain study subjects on isocaloric diets with a stable exercise routine with no new medications being allowed to minimize influences on period effects.

[0114] Furthermore, the present inventors have used a low total daily dose of mifepristone to avoid drug-induced adrenal insufficiency or other side effects. Additionally, mifepristone was administered four (4) times daily based upon the hypothesis that such a regimen would take advantage of a differentially higher exposure of the pituitary gland compared to the omentum and liver. The present inventors postulated that the first pass effect of an oral dose of mifepristone would provide a larger mifepristone “dose” to the pre-hepatic and hepatic tissues, and reduce 11- β HSD1 activity, before hepatic metabolism resulted in a smaller circulating level that would be recognized at the pituitary gland and stimulate ACTH. This objective was achieved as evidenced by the fact that UFC values were normal or within the expected stress range. Presumably, that stress range of UFC was sufficient to be blocked by the GR antagonist, so that peripheral cortisol effects (apart from pre-hepatic effects) were normal.

Weight Loss Therapy

[0115] The present invention also provides a method for promoting weight loss by administering one or more glucocorticoid receptor antagonists effective to reduce body weight of a patient without activating the hypothalamic-pituitary-adrenal axis or with minimal activation of the hypothalamic-pituitary-adrenal axis as shown by 24-hour urine cortisol less than three times normal or mean 24-hour serum cortisol less than 200% (twice normal) above the normal range.

[0116] In essence, weight loss in a patient is the concomitant result of amelioration of insulin resistance and a result of decreased 11 β HSD1 regeneration of cortisol from cortisone.

[0117] In this additional method, the same compounds and dosages apply as discussed above for

the amelioration of insulin disorders.

Neutralizing Anti-Progestin Effects of Glucocorticoid Receptor Antagonists To Enhance Selectivity of the Antagonists for Amelioration of Insulin Disorders

[0118] Another important aspect of the present invention involves the administration of progestin agonists to enhance the selectivity of the glucocorticoid receptor antagonists for only the glucocorticoid receptors. This is particularly important where the glucocorticoid receptor antagonists used are mifepristone or metapristone or any GR antagonist that also antagonizes progesterone action.

[0119] Such antiprogestin activity may impair folliculogenesis and estrogen production, or inhibit the mid-cycle LH surge. This disrupts the normal menstrual cycle so that estrogen production may be reduced dramatically and menses may occur at unexpected times or not at all. An important effect of chronic reductions in estrogen production is a decrease in bone mineral density, which, over time, may lead to osteoporosis or fractures. To prevent this problem, estrogen agonists may be given with the progestin agonists to provide adequate estrogen levels.

[0120] Intact progesterone itself may be used as the human progestin agonist, any 11beta-substituted progesterone analog may also be used. For example, any of the compounds of U.S. Pat. No. 5,073,548 may be used with and in conjunction with this aspect of the present invention. Each and all of U.S. Pat. Nos. 5,073,548; 6,756,388 and 6,608,203 are incorporated herein in the entirety by reference as disclosing estrogen receptor (ER) agonists which may be used in this aspect of the present invention.

[0121] Further, the progestin receptor agonists may be administered before or with the GR antagonists. When the components are administered simultaneously, they may be administered separately or as a composition either as a binary composition or with added excipients. The progestin agonists, much like progesterone itself, may be administered in amounts of from about 50-750 mg/daily, and more preferably from about 200-400 mg/daily.

[0122] Dosages of estrogen receptor (ER) agonists used in accordance with the present invention should be below or above those used for postmenopausal women. For example, daily doses of estradiol in the range of about 0.25 to 2.0 mg may be used, and preferably from about 0.5 to 1 mg. Daily dosages of ethinyl estradiol in the range of about 10 to 50 mg may be used, and preferably from about 20 to 35 mg. The above dosages may be modified by a treating physician in accordance with patient indications and professional judgment. Regular birth control pills may also be used in similar dosages.

[0123] Thus, the present invention specifically contemplates such compositions containing both one or more progestin receptor agonists, with or without one or more estrogen agonists, in addition to one or more glucocorticoid receptor antagonists.

[0124] Generally, about 2-20 times on a molar basis of one or more progestin receptor agonists (PRAs) per mole of one or more GR antagonists (GRAs) are used either with prior use of the one or more progestin agonists or use of the PR agonists concomitantly with the one or more GR antagonists. More preferably, however, the ratio of PRAs/GRAs is from about 5-10, regardless of whether the use of PRAs is prior to or with the GRAs. These molar ratios may be used routinely by those in the art to determine either separate dosages or dosages when used in combination as in binary pharmaceutical compositions

Measuring Bone Density

[0125] As noted above, the use of ER agonists in the present invention serves to maintain or increase bone density in reproductive-aged women. A treating physician may use a standard bone density test to establish the need for the use of ER agonists as disclosed herein. The most widely recognized bone mineral density (BMD) test is the dual energy X-ray absorptiometry (DXA) test. A BMD test measures a patient's bone mineral density and compares that to an established norm or standard to provide a score.

[0126] Briefly, a BMD score is "normal" if measured bone density is within 1 standard deviation

(SD) (+1 or -1) of the young adult mean; the score indicates “osteopenia” if it is -1 to -2.5 SD; the score indicates “osteoporosis” if it is -2.5 SD or lower; and the score indicates “severe (established) osteoporosis” if it is -2.5 SD or lower with one or more prior osteoporotic fractures. [0127] A treating physician may rely on these bone mineral density standards as measured by the DXA test to determine whether treatment to increase bone mineral density is advisable.

Pharmaceutical Compositions

[0128] Pharmaceutical compositions may be formulated as single, binary or ternary compositions depending on whether ER agonists and/or progestin agonists are added to the one or more GR antagonists as disclosed herein. Further, any of these compositions may be formulated using any of the excipients disclosed in the U.S. patents incorporated by reference in the entirety with this application. These excipients are all necessarily pharmaceutically-acceptable.

[0129] For example, any of the excipient formulations in U.S. Pat. No. 10,238,659 may be used with one or more GR antagonists as disclosed herein in accordance with the present invention. However, certain excipient formulations of this patent may be more preferable than others based on solubility characteristics of the particular one or more GR antagonists employed in the formulation. Determining the optimal excipient formulation is within the ordinary skill of the artisan. U.S. Pat. No. 10,238,659, as noted above, is incorporated herein in the entirety by reference. While any method of administration may be employed, oral administration is one of the preferred methods of administration. Hence, the one or more GR antagonists described herein may be administered in the form of tablets or capsules with the one or more GR antagonists being included in amounts of from about 25 to 75 mg per single dose. Such doses are typically administered several times daily as described herein to improve insulin sensitivity and/or lose weight, for example.

[0130] All GR antagonists described herein are all known, and can either be purchased commercially or may be synthesized by known synthetic methodologies. This specifically includes ORIC-101, Relacorilant, Miricorilant and Exicorilant. Finally, any of the methods and compositions described above may include administration of a mineralocorticoid receptor (MR) antagonist, such as spironolactone or eplerenone, to antagonize the MR activation caused by excess cortisol, which may be responsible for increased blood pressure. See US 2015/0284376, which is incorporated herein in the entirety, and also Kolkhof P et al. *J. Endocrinol.* 2017 July; 234(1): Y125-T140.

Claims

1. A method of ameliorating a disorder of insulin sensitivity, which comprises administering an amount of one or more glucocorticoid receptor antagonists to a patient in need thereof effective to ameliorate said disorder, and which method is effected under non-rescue conditions.
2. The method of claim 1, wherein said disorder of insulin insensitivity is selected from the group consisting of diabetes, prediabetes, obesity, lipodystrophy, hypertriglyceridemia, non-alcoholic fatty liver disease and growth hormone excess states.
3. The method of claim 1 wherein said non-rescue conditions are a non-pathological cause of hypercortisolism.
4. The method of claim 1, wherein said one or more glucocorticoid receptor antagonists are steroidal.
5. The method of claim 1, wherein said one or more glucocorticoid receptor antagonists is one antagonist, mifepristone.
7. The method of claim 1, wherein said one or more glucocorticoid antagonists is at least one of mifepristone, metapristone or ORIC-101.
8. The method of claim 1, wherein said one or more glucocorticoid receptor antagonists are non-steroidal.
9. The method of claim 8, wherein the non-steroidal GR antagonist is Relacorilant, Miricorilant or

Exicorilant.

10. The method of claim 1, wherein said one or more non-steroidal compound is a single compound, [bis(4-N,N-dimethylaminophenyl)(2-chloro-5-nitrophenyl) methane].

11. The method of claim 1, wherein said one or more glucocorticoid receptor antagonists are administered in an amount of about 0.5 mg to about 100 mg per dose to a human patient one or more times per day.

12. The method of claim 1, wherein said administration is oral, transdermal, topical, intra-vaginal, rectal, by suppository, or by depot.

13. The method of claim 1, which improves insulin sensitivity in hepatic and adipose tissue.

14. The method of claim 1, which further comprises measuring a cortisol level or biochemical surrogate cortical marker of said patient after a steady-state administration of said one or more glucocorticoid receptor antagonists to determine a preferred dose of said one or more glucocorticoid receptor antagonists for continued administration.

15. The method of claim 1, which is effected without excessive activation of the hypothalamic-pituitary-adrenal axis such that 24-hour serum or urine cortisol levels do not exceed more than about two or three times the upper limit of normal, respectively.

16. A method of promoting weight loss in a patient, which comprises administering to a patient an amount of one or more glucocorticoid receptor antagonists effective to promote weight loss without excessive activation of the hypothalamic-pituitary-adrenal axis, such that 24-hour urine free cortisol or 24-h hour serum cortisol levels are no more than about three or two times the upper limit of normal, respectively.

17. The method of claim 16, wherein said one or more glucocorticoid receptor antagonists comprise mifepristone or an in vivo metabolite thereof, or an in vivo metabolite of the analog.

18. The method of claim 16, wherein said one or more glucocorticoid receptor antagonist is the single compound mifepristone.

19. The method of claim 1, which further comprises, in women, enhancing selectivity of the at least one glucocorticoid antagonist for binding with glucocorticoid receptors rather than progestin receptors in a patient by administering one or more progestin receptor agonists either before or with said at least one glucocorticoid receptor antagonist.

20. The method of claim 19, wherein both of said one or more progestin receptor antagonists, and said at least one glucocorticoid receptor antagonists are administered together in a composition.

21. The method of claim 20, wherein the composition comprises at least one of progesterone or an 11 β -substituted analog thereof and at least one of mifepristone or metapristone.

22. The method of claim 1, which further comprises administering a mineralocorticoid receptor antagonist to antagonize mineralocorticoid activation caused by excess cortisol produced by an HPA axis feedback loop when the GR is blocked.

23. The method of claim 22, wherein said mineralocorticoid receptor antagonist is at least one of spironolactone or eplerenone.

24. A method of decreasing insulin resistance in a human with normal cortisol regulation, which comprises administering an amount of one or more glucocorticoid receptor antagonists effective to decrease said insulin resistance, and which method is conducted under non-rescue conditions.
